Responses to Peer Review Comments

Conference:
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Paper Title: Coordinated Optimization of Emergency Supply Distribution and Multi-Type Personnel Evacuation under Hazardous-Material Vehicle Incidents on Urban Expressways: A Case Study of Beijing

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Date: 2026-05-12

Please read the following instructions carefully and complete this form with responses to the reviewer comments.

i. Individual comment from reviewers should be organized in order and followed by clear and concise response from authors.
ii. Changes should be highlighted within the revised manuscript and the revision should be described in this form.
iii. If the authors feel unable to address a comment or implement a suggestion, please state the reason in this form.
iv. The completed form should be uploaded with the revised manuscript by the agreed deadline.

Response to Comments from Reviewer 1

Overall Comment:

The topic presented in the paper is valid and important. In the reviewed paper, the Authors presented the problem of coordinated optimization of emergency supply distribution and multi-type personnel evacuation under hazardous-material vehicle incidents on urban expressways - as a case study of Beijing. Using a hypothetical chlorine release at Beijing's Sixth Ring Road as a scenario, this study develops an integrated emergency-response framework. The Authors used the Gaussian plume model and ERPG thresholds to delineate three-tier hazard zones and co-optimized a truck-drone logistics network and a multi-type evacuation fleet encompassing five modes and three vulnerability classes. Results show that incorporating shelter-in-place demand increases supply tasks from 40 to 70. Under lightly damaged road conditions, the truck-only scheme outperforms the truck-drone scheme, highlighting the relay delivery's

applicability boundary. ERPG-based zoning reduces the active evacuation population from 235,000 to 13,800. Vulnerable groups are 100% evacuated, but critically injured victims achieve only a 2.8% evacuation rate, exposing a structural bottleneck in medical fleets. The paper is well-structured and well-organized. The paper content is suitable for the conference scope.

Response:

We thank Reviewer 1 for the positive evaluation of our work and for the careful summary of our contributions. We are encouraged that the reviewer found the topic valid and important, the framework integrated, and the paper well-structured and suitable for the conference scope. In the revised manuscript we have tightened the writing throughout the abstract, introduction, literature review, research framework, results, discussion, and conclusions for clarity and conciseness in line with the conference page limit, and have made several targeted improvements in response to Reviewer 2's specific comments. All paragraphs modified relative to the original submission are marked in yellow highlight in the revised manuscript for ease of identification.

Comment 1:

(No specific comments - see Overall Comment above.)

Response:

(See Overall Response above.)

Comment 2:

(No specific comments - see Overall Comment above.)

Response:

(See Overall Response above.)

Comment 3:

(No specific comments - see Overall Comment above.)

Response:

(See Overall Response above.)

Response to Comments from Reviewer 2

Overall Comment:

This paper proposes a coordinated emergency-response framework for hazardous-material vehicle incidents on urban expressways, integrating emergency supply distribution and multi-type personnel evacuation under a chlorine-release scenario on Beijing's Sixth Ring Road. Using Gaussian plume

modelling, ERPG-based zoning, a truck-drone logistics network, and multimodal evacuation dispatch, the paper shows that differentiated zoning can substantially reduce the active evacuation population, that vulnerable groups can be effectively protected, and that medical evacuation capacity is the main system bottleneck. The paper is relevant to the conference scope and addresses an important problem in emergency transportation management. Its main strengths are the integrated perspective, the realistic urban case setting, and the connection between optimization results and resilience-oriented policy implications. However, several revisions are needed.

Response:

We thank Reviewer 2 for the detailed and constructive overall assessment, and for recognizing the integrated perspective, realistic urban case setting, and the connection between optimization results and resilience-oriented policy implications as the paper's main strengths. We have carefully addressed each of the four specific revision points raised by the reviewer; the corresponding revisions are highlighted in yellow in the revised manuscript and are described in detail in our responses to Comments 1 through 4 below.

Comment 1:

Result consistency and methodological transparency (1). In Section 4.3, the text reports that the truck-only scheme completes 64 tasks and the truck-drone scheme completes 62 tasks, while Table 2 reports 61 and 60, respectively.

Response:

We thank the reviewer for catching this inconsistency. The values in Table 2 (truck-only: 61 completed; truck-UAV: 60 completed; 4 relay deliveries) are the correct outputs of the final optimization run; the 64/62/34 figures in the original Section 4.3 narrative were carried over from an earlier draft. In the revised manuscript we have reconciled the narrative with Table 2 throughout the paper: Section 4.3 now reads "completes 61 tasks (87.1%) ... 4 relay tasks vs. 0 in the truck-only baseline ... completes 1 fewer task"; Section 5.3 now reads "relay deliveries from 0 to 4 tasks ... 1 fewer task is completed"; and the Conclusions now report "87.1% supply completion with 100% on-time life-rescue". The qualitative findings of Section 4.3 - that shelter-in-place demand expands the workload to 70 tasks, that the truck-only scheme outperforms the truck-drone scheme on a lightly damaged network, that the drone-only scheme is infeasible due to payload limits, and that the 6 unfinished tasks reflect supply-point inventory limits - are unaffected and remain consistent with Table 2.

Comment 2:

Result consistency and methodological transparency (2). The optimization procedures are described only at a high level. The paper should more clearly explain the objective-function components, weight settings, relay-trigger rule, unmet-demand penalty, and how the greedy + 2-opt algorithm is implemented. Several key parameters also require justification, including the 5 kg/min

chlorine release rate, the assumed 2% k=1 critically injured share in the ERPG-1 zone, fleet sizes, and inventory capacities.

Response:

We agree that more transparency on the optimization model and parameters would strengthen the paper, and we appreciate the reviewer's guidance. Within the strict 8-page conference limit (already filled with case-study figures, the data-source table, and the supply-scheme comparison table), it is not feasible to include the full algorithmic specification and complete parameter tables. We have nevertheless made the following targeted improvements to Section 3.2 (highlighted in yellow): (i) the objective function is now explicitly stated as a weighted sum of makespan, time-window penalties, and unmet demand, with the unmet-demand penalty applied per missed task and weighted by priority class; (ii) the relay-trigger rule is now stated as "relay is triggered when the destination lies inside the closure zone or the direct truck travel time exceeds a threshold"; (iii) the two-stage greedy + 2-opt structure is retained and clarified; and (iv) parameter justifications are added directly in the text: the 5 kg/min release rate is consistent with the order of magnitude of historical tanker-incident scenarios (the 2005 Huai'an event cited in the introduction); the 2% k=1 share in ERPG-1 is a conservative upper bound used to stress-test medical capacity (refined in future work via concentration-dependent injury rates, e.g., 5% / 1% / 0.2% in ERPG-3 / 2 / 1); and the fleet sizes (15 ambulances, 3 helicopters, buses, private cars, metro trains) and supply-point inventory are calibrated to the actual emergency-resource inventory of the study area. A full mathematical specification, complete parameter table, and the implementation code are being prepared for the extended journal version of this work.

Comment 3:

Result consistency and methodological transparency (3). The resilience discussion is insightful, but the paper currently uses "resilience" mainly as an interpretive concept; the indicators and evaluation logic should be stated more explicitly.

Response:

We have made the resilience-assessment logic explicit in the revised research framework (Section 2.3, Layer 3; highlighted in yellow). The Layer 3 description now reads: "Layer 3 assesses resilience along three dimensions with explicit indicators: structural resilience (closure length and ratio within the influence zone), functional resilience (supply-task completion rate, on-time life-rescue rate, evacuation completion rate by vulnerability class, and cumulative risk exposure), and adaptive resilience (marginal improvement from technology- or policy-driven interventions, such as truck-drone relay or fleet expansion)." The asymmetric "high-supply / low-evacuation" resilience pattern reported in Section 5.1 follows directly from comparing these indicators across the supply and evacuation chains. We agree with the reviewer that a fully formalized resilience metric (e.g., area-under-the-recovery-curve indicators) would further sharpen the analysis, and we plan to develop such a metric in the journal extension of this work.

Comment 4:

Result consistency and methodological transparency (4). Because conclusions are based on one hypothetical scenario, the paper would benefit from limited additional sensitivity analysis or stronger discussion of scenario dependence.

Response:

We have strengthened both the sensitivity analysis and the discussion of scenario dependence in the revised manuscript. Section 4.2 reports four wind-direction scenarios (225 degrees baseline; 0 degrees; 90 degrees; and 225 degrees at 8 m/s), confirming that wind direction decisively shifts which population centers and infrastructure clusters fall within the hazard zones and underscoring the need for real-time meteorological integration. Section 4.4 reports fleet-composition sensitivity: removing helicopters lowers the evacuation rate to 40.2% and raises cumulative risk by 6.4% (85,694 -> 91,153), while removing the metro lowers it to 38.8%, quantifying the marginal contributions of dedicated-medical and high-capacity public-transport assets. In Section 5.3 we have made the applicability boundary of truck-drone relay explicit: under the present lightly damaged network (closure ratio 19.1%) relay is not beneficial, but its advantage is expected to grow as damage intensifies. In the Conclusions (highlighted in yellow) we have added the explicit caveat that "these conclusions are conditional on the single hypothetical scenario" and pointed to a systematic multi-scenario sensitivity study covering release rate, atmospheric stability, and time-of-day demand as the next step for the journal extension.

Response to Comments from Reviewer 3

Overall Comment:

(Not applicable - only two reviewers for this paper.)

Response:

(Not applicable.)

Comment 1:

(Not applicable.)

Response:

(Not applicable.)

Comment 2:

(Not applicable.)

Response:

(Not applicable.)

Comment 3:

(Not applicable.)

Response:

(Not applicable.)